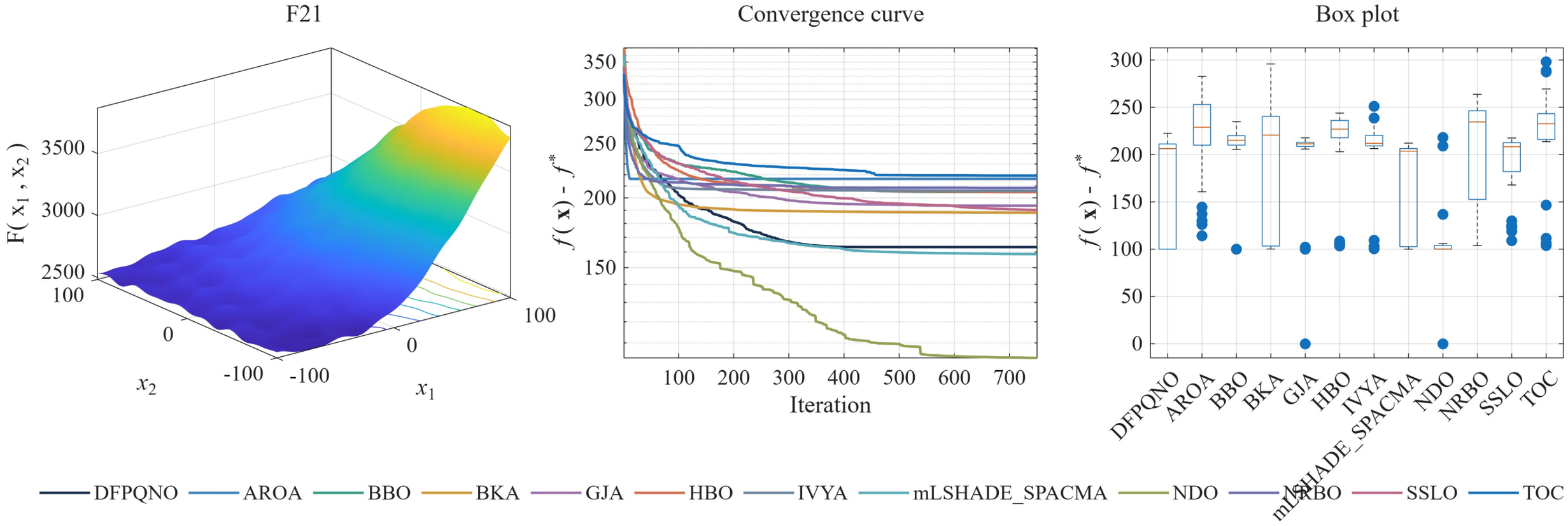

F21
F(x1 , x2)
3500
3000
2500
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
350
300
250
200
150
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
300
250
200
150
100
50
0
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC